

Airplane Flying Handbook (FAA-H-8083-3C)

Chapter 2: Ground Operations

Introduction

Experienced pilots place a strong emphasis on ground operations as this is where safe flight begins and ends. They know that hasty ground operations diminish their margin of safety. A smart pilot takes advantage of this phase of flight to assess various factors including the regulatory requirements, the pilot's readiness for pilot-in-command (PIC) responsibilities, the airplane's condition, the flight environment, and any external pressures that could lead to inadequate control of risk.

Flying an airplane presents many new responsibilities not required for other forms of transportation. Focus is often placed on the flying portion itself with less emphasis placed on ground operations. However pilots need to allow time for flight preparation. Situational awareness begins during preparation and only ends when the airplane is safely and securely returned to its tie-down or hangar, or if a decision is made not to go.

This chapter covers the essential elements for the regulatory basis of flight including:

1. An airplane's airworthiness requirements,
2. Important inspection items when conducting a preflight visual inspection,
3. Managing risk and resources, and
4. Proper and effective airplane surface movements using the AFM/POH and airplane checklists.

Preflight Assessment of the Aircraft

The visual preflight assessment mitigates airplane flight hazards. The preflight assessment ensures that any aircraft flown meets regulatory airworthiness standards and is in a safe mechanical condition prior to flight. Per 14 CFR part 3, section 3.5(a), the term "airworthy" means that the aircraft conforms to its type design and is in condition for safe operation. The owner/operator is primarily responsible for maintenance, but in accordance with 14 CFR part 91, section 91.7(a) and (b) no person may operate a civil aircraft unless it is in an airworthy condition and the pilot in command of a civil aircraft is responsible for determining whether the aircraft is in condition for safe flight. The pilot's inspection should involve the following:

1. Inspecting the airplane's airworthiness status.
2. Following the AFM/POH to determine the required items for visual inspection. [Figures 2-1, 2-2, 2-3].

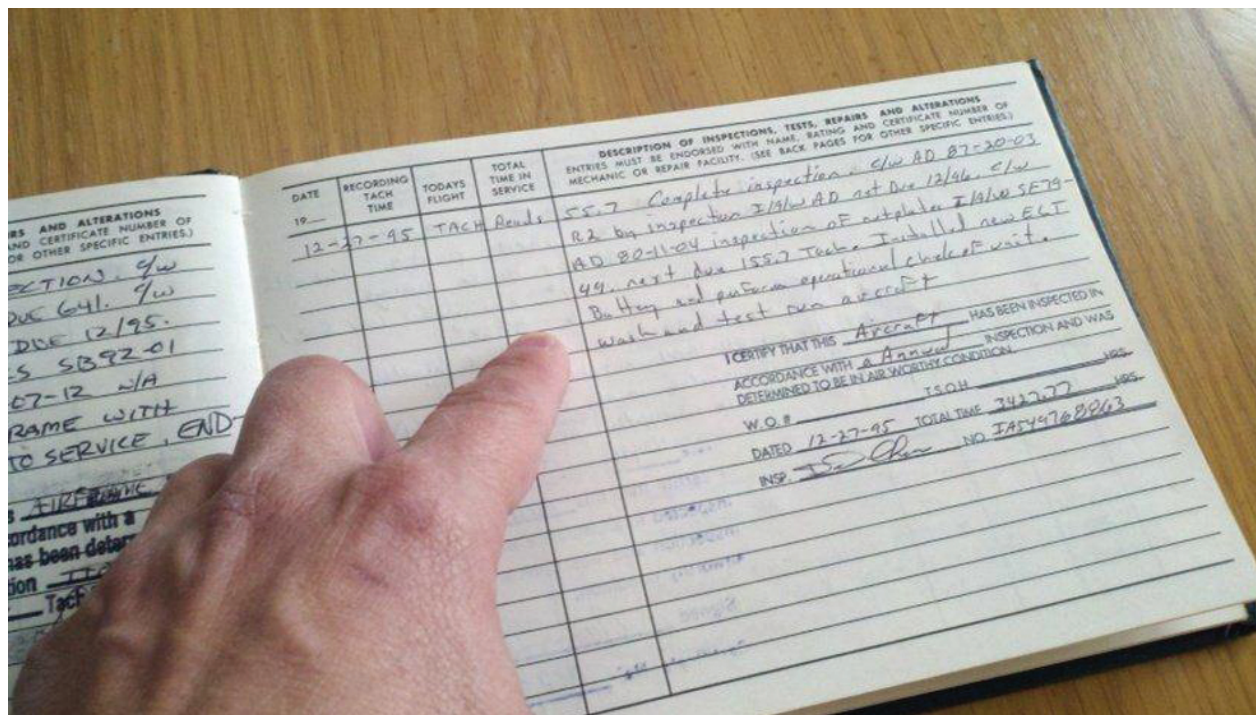


Figure 2-1. Pilots should view the aircraft's maintenance logbook prior to flight to ensure the aircraft is safe to fly.



Figure 2-2. A visual inspection of the aircraft before flight is an important step in mitigating airplane flight hazards.

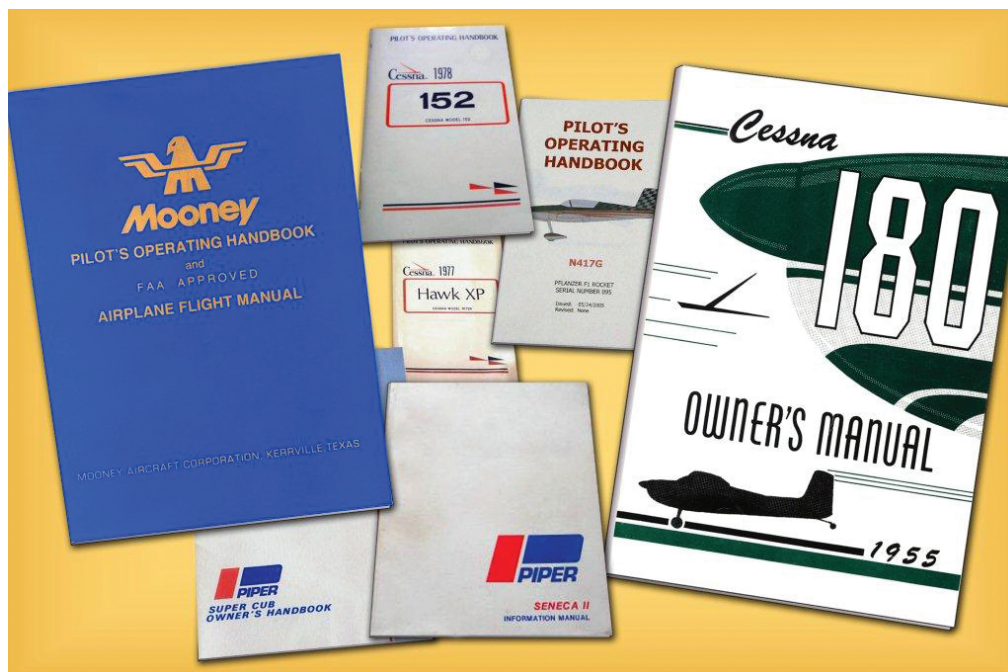


Figure 2-3. Airplane Flight Manuals (AFM) and the Pilot Operating Handbook (POH) for each individual aircraft explain the required items for inspection.

Each airplane has a set of logbooks that include airframe and engine, and in some cases, propeller and appliance logbooks, which are used to record maintenance, alteration, and inspections performed on a specific airframe, engine, propeller, or appliance. It is important that the logbooks be kept accurate, secure, and available for inspection. Airplane logbooks are not normally kept in the airplane. It should be a matter of procedure by the pilot to inspect the airplane logbooks or a summary of the airworthy status prior to flight to ensure that the airplane records of maintenance, alteration, and inspections are current and correct. [Figure 2-4] The following is required:

- Annual inspection within the preceding 12 calendar months (Title 14 of the Code of Federal Regulations (14 CFR) part 91, section 91.409(a))
- 100-hour inspection, if the aircraft is operated for hire (14 CFR part 91, section 91.409(b))
- Transponder certification within the preceding 24 calendar months (14 CFR part 91, section 91.413)
- Static system and encoder certification, within the preceding 24 calendar months, required for instrument flight rules (IFR) flight in controlled airspace (14 CFR part 91, section 91.411)
- 30-day VHF omnidirectional range (VOR) equipment check when using the VOR system of radio navigation for IFR flight (14 CFR part 91, section 91.171)
- Emergency locator transmitter (ELT) inspection within the last 12 months (14 CFR part 91, section 91.207(d))
- ELT battery due (14 CFR part 91, section 91.207(c))
- Current status of life limited parts per Type Certificate Data Sheets (TCDS) (14 CFR part 91, section 91.417)
- Status, compliance, logbook entries for airworthiness directives (ADs) (14 CFR part 91, section 91.417(a)(2)(v))
- Federal Aviation Administration (FAA) Form 337, Major Repair or Alteration (14 CFR part 91, section 91.417)
- Inoperative equipment (14 CFR part 91, section 91.213)

Airplane Documents

- ☒ Original Airworthiness Certificate (FAR 91.203)
- ☒ Original Registration Certificate (FAR 91.203)
- ☒ Radio Station License for flights outside the United States or airplanes greater than 12,500 pounds (FCC Rule)
- ☒ Operating Limitations, which may be in the form of an FAA-approved Airplane Flight Manual and/or Pilot's Operating Handbook (AFM/POH), Placards, instrument markings, or any combination thereof (FAR 91.9)
- ☒ Official Weight and Balance
- ☒ Compass Deviation Card (FAR 23.1547)
- ☒ External Data Plate (FAR 45.11)

Airplane Inspections

- ☒ Annual—12 months: Current Due 12-31-2012
- ☒ 100 hour—Next Due Hobbs 3245.7
- ☒ VOR—30 Day Next Due 7-30-2011
- ☒ Altimeter—24 months: Due 1-31-2013
- ☒ Static System and Encoder—24 months: Due 1-31-2013

Required Equipment

VFR (Day)—FAR 91.205b

- ☒ Altimeter
- ☒ Tachometer for each engine
- ☒ Oil temperature gauge
- ☒ Manifold pressure gauge
- ☒ Airspeed Indicator
- ☒ Temperature Gauge
- ☒ Oil pressure Gauge
- ☒ Fuel Level Indicator
- ☒ Landing Gear Position Indicator
- ☒ Anti-Collision Light
- ☒ Magnetic Compass
- ☒ ELT (Except as required by FAR 91.207(d))
- ☒ Safety Belts with Harness

VFR Night—FAR 91.205c

- ☒ Fuses
- ☒ Landing Light
- ☒ Anti-collision Light
- ☒ Position Light
- ☒ Source of Electrical Power

IFR—FAR 91.205d

- ☒ Generator

Figure 2-4. A sample airworthiness checklist used by pilots to inspect an aircraft.